

EXPERIMENT – 6

Snort IDS/IPS to Detect and Prevent Real-Time Threats Using TryHackMe Platform

Aim:

To understand and demonstrate the working of Snort as an Intrusion Detection System (IDS) and Intrusion Prevention System (IPS) by detecting real-time network threats using the TryHackMe platform.

Tools / Platform Required:

- Web Browser (Chrome / Firefox)
- Internet Connection
- TryHackMe Account (Already Created)
- TryHackMe AttackBox (Built-in Linux Environment)
- Snort IDS/IPS Tool (Pre-installed in AttackBox)

Procedure:

Step 1: Login to TryHackMe

1. Open browser and go to: <https://tryhackme.com>
2. Login using registered credentials.

Step 2: Open Snort Room

1. In the search bar, type Snort.
2. Select a room related to Snort IDS/IPS.
3. Click Join Room.

Step 3: Start the AttackBox

1. Click Start AttackBox.
2. Wait for the Linux environment to load.
3. Open the terminal inside AttackBox.

Step 4: Verify Snort Installation

1. In the terminal, check Snort version:
2. `snort --version`
3. Ensure Snort is successfully installed.

Step 5: Run Snort in IDS Mode

1. Start Snort in console mode:
`sudo snort -A console -q -c /etc/snort/snort.conf -i eth0`
2. Snort begins monitoring network traffic in real time.

Step 6: Detect Network Threats

1. Perform network scanning or simulated attacks as instructed in the TryHackMe room.
2. Observe alerts generated by Snort.
3. Analyze alerts indicating suspicious or malicious activity.

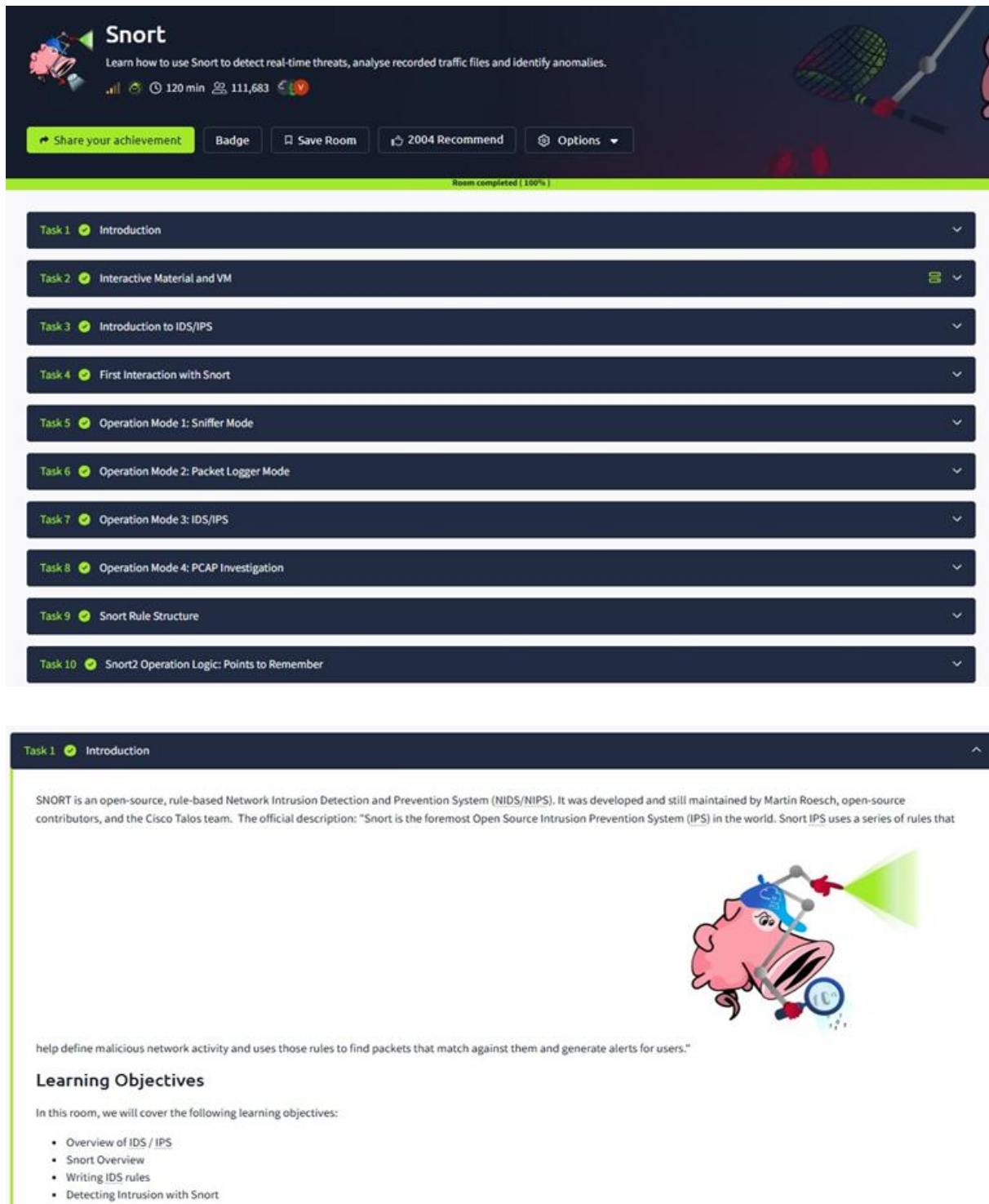
Step 7: Configure Snort as IPS (Optional)

1. Run Snort in inline (IPS) mode:
`sudo snort -Q --daq afpacket -c /etc/snort/snort.conf -i eth0`
2. Configure rules to drop malicious packets.
3. Observe blocked traffic in real time.

Step 8: Analyze Alerts

1. Review generated alerts on the terminal.
2. Identify attack types such as port scanning, brute force, or suspicious packets.

Output:



The screenshot displays the Snort Cyber Range interface. At the top, the 'Snort' logo is accompanied by the text 'Learn how to use Snort to detect real-time threats, analyse recorded traffic files and identify anomalies.' Below this, a progress bar indicates 'Room completed (100%)'. A list of tasks is shown, all marked as completed with green checkmarks:

- Task 1: Introduction
- Task 2: Interactive Material and VM
- Task 3: Introduction to IDS/IPS
- Task 4: First Interaction with Snort
- Task 5: Operation Mode 1: Sniffer Mode
- Task 6: Operation Mode 2: Packet Logger Mode
- Task 7: Operation Mode 3: IDS/IPS
- Task 8: Operation Mode 4: PCAP Investigation
- Task 9: Snort Rule Structure
- Task 10: Snort2 Operation Logic: Points to Remember

The 'Task 1: Introduction' section is expanded, showing the following text:

SNORT is an open-source, rule-based Network Intrusion Detection and Prevention System (NIDS/NIPS). It was developed and still maintained by Martin Roesch, open-source contributors, and the Cisco Talos team. The official description: "Snort is the foremost Open Source Intrusion Prevention System (IPS) in the world. Snort IPS uses a series of rules that help define malicious network activity and uses those rules to find packets that match against them and generate alerts for users."

Learning Objectives

In this room, we will cover the following learning objectives:

- Overview of IDS / IPS
- Snort Overview
- Writing IDS rules
- Detecting Intrusion with Snort

A cartoon illustration of a pink pig wearing a blue helmet and holding a magnifying glass is visible on the right side of the task details.

Prerequisites

This room expects you to be familiar with basic Linux command-line functionalities like general system navigation and Network fundamentals (ports, protocols, and traffic data). The room aims to encourage you to start working with Snort to analyse live and captured traffic. Before joining this room, explore the following rooms:

- [Network Fundamentals Module](#).
- [Linux Fundamentals rooms \(1 2 3 \)](#).

Answer the questions below

Read the task above.

No answer needed

✓ Correct Answer

Task 2 ✓ Interactive Material and VM

Answer the questions below

Navigate to the Task-Exercises folder and run the command `./easy.sh` and write the output

Too Easy!

✓ Correct Answer

Task 3 ✓ Introduction to IDS/IPS

Answer the questions below

Which IDS or IPS type can help you stop the threats on a local machine?

HIPS

✓ Correct Answer

Which IDS or IPS type can help you detect threats on a local network?

NIDS

✓ Correct Answer

Which IDS or IPS type can help you detect the threats on a local machine?

HIDS

✓ Correct Answer

Which IDS or IPS type can help you stop the threats on a local network?

NIPS

✓ Correct Answer

Which described solution works by detecting anomalies in the network?

NBA

✓ Correct Answer

According to the official description of the snort, what kind of NIPS is it?

full-blown

✓ Correct Answer

NBA training period is also known as ...

baselining

✓ Correct Answer

Task 4 ✓ First Interaction with Snort

The First Interaction with Snort

First, let's verify Snort is installed. The following command will show you the instance version.

```
version check

user@ubuntu$ snort -V

,,_  -*> Snort! <*-
o"  )~ Version 2.9.7.0 GRE (Build XXXXXX)
'''  By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
      Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved.
      Copyright (C) 1998-2013 Sourcefire, Inc., et al.
      Using libpcap version 1.9.1 (with TPACKET_V3)
      Using PCRE version: 8.39 2016-06-14
      Using ZLIB version: 1.2.11
```

Answer the questions below

Run the Snort instance and check the build number.

✓ Correct Answer



Test the current instance with "/etc/snort/snort.conf" file and check how many rules are loaded with the current build.

✓ Correct Answer



Test the current instance with "/etc/snort/snortv2.conf" file and check how many rules are loaded with the current build.

✓ Correct Answer



Task 5 Operation Mode 1: Sniffer Mode



Let's run Snort in Sniffer Mode.

Like tcpdump, Snort has various flags that allow for viewing different data about the packet it is ingesting.

Sniffer mode parameters are explained in the table below;

Parameter	Description
-v	Verbose. Display the <u>TCP</u> /IP output in the console.
-d	Display the packet data (payload).
-e	Display the link-layer (<u>TCP</u> /IP/UDP/ICMP) headers.
-X	Display the full packet details in HEX.
-i	This parameter helps define a specific network interface to listen to or sniff. Once you have multiple interfaces, you can choose a specific interface to sniff.

Note that you can use the parameters both in combined and separated form as follows;

- snort -v
- snort -vd
- snort -de
- snort -v -d -e
- snort -X

Make sure you understand and practice each parameter with different types of traffic and discover your favourite combination.

Answer the questions below

You can practice the parameter combinations by using the traffic-generator script.

✓ Correct Answer

Task 6 Operation Mode 2: Packet Logger Mode

```
Snort -dvr logname.log -n 10
```

Please use the following resources to understand how the BPF works and its use.

- https://en.wikipedia.org/wiki/Berkeley_Packet_Filter
- <https://biot.com/capstats/bpf.html>
- <https://www.tcpdump.org/manpages/tcpdump.1.html>

Now, use the attached VM and navigate to the Task-Exercises/Exercise-Files/TASK-6 folder to answer the questions!

Answer the questions below

Investigate the traffic with the default configuration file with **ASCII mode**.

```
sudo snort -dev -K ASCII -l .
```

Execute the traffic generator script and choose "**TASK-6 Exercise**". Wait until the traffic ends, then stop the Snort instance. Now analyse the output summary and answer the question.

```
sudo ./traffic-generator.sh
```

Now, you should have the logs in the current directory. Navigate to folder "**145.254.160.237**". What is the source port used to connect port 53?

✓ Correct Answer



Use **snort.log.1640048004**

Read the snort.log file with Snort; what is the IP ID of the 10th packet?

```
snort -r snort.log.1640048004 -n 10
```

✓ Correct Answer



Read the "**snort.log.1640048004**" file with Snort; what is the referer of the 4th packet?

✓ Correct Answer



Read the "**snort.log.1640048004**" file with Snort; what is the Ack number of the 8th packet?

✓ Correct Answer

Read the "**snort.log.1640048004**" file with Snort; what is the number of the "**TCP port 80**" packets?

✓ Correct Answer



Task 7 Operation Mode 3: IDS/IPS

Let's run Snort in IDS/IPS Mode

NIDS mode parameters are explained in the table below;

Parameter	Description
-c	Defining the configuration file.
-T	Testing the configuration file.
-N	Disable logging.
-D	Background mode.
-A	Alert modes; Full: Full alert mode, providing all possible information about the alert. This mode is also the default; once you use -A and don't specify a mode, Snort defaults to this mode. Fast mode displays the alert message, timestamp, source and destination IP addresses , along with port numbers. Console: Provides fast style alerts on the console screen. cmg: CMG style, basic header details with payload in hex and text format. None: Disabling alerting.

Answer the questions below

Investigate the traffic with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l .
```

Execute the traffic generator script and choose "TASK-7 Exercise". Wait until the traffic stops, then stop the Snort instance. Now analyse the output summary and answer the question.

```
sudo ./traffic-generator.sh
```

What is the number of the detected HTTP GET methods?

✓ Correct Answer



You can practice the rest of the parameters by using the traffic-generator script.

✓ Correct Answer

Task 8 Operation Mode 4: PCAP Investigation

Answer the questions below

Investigate the **mx-1.pcap** file with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l . -r mx-1.pcap
```

What is the number of the generated alerts?

✓ Correct Answer

Keep reading the output. How many TCP Segments are Queued?

✓ Correct Answer

Keep reading the output. How many "HTTP response headers" were extracted?

✓ Correct Answer

Investigate the **mx-1.pcap** file with the **second** configuration file.

```
sudo snort -c /etc/snort/snortv2.conf -A full -l . -r mx-1.pcap
```

What is the number of the generated alerts?

✓ Correct Answer

Investigate the **mx-2.pcap** file with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l . -r mx-2.pcap
```

What is the number of the generated alerts?

✓ Correct Answer



Keep reading the output. What is the number of the detected TCP packets?

✓ Correct Answer

Investigate the **mx-2.pcap** and **mx-3.pcap** files with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l . --pcap-list="mx-2.pcap mx-3.pcap"
```

What is the number of the generated alerts?

✓ Correct Answer

Task 9 Snort Rule Structure

Answer the questions below

Use "task9.pcap". Write a rule to filter IP ID "35369" and run it against the given pcap file. What is the request name of the detected packet? You may use this command: "snort -c local.rules -A full -l . -r task9.pcap"

✓ Correct Answer



Clear the previous alert file and comment out the old rules. Create a rule to filter packets with **Syn** flag and run it against the given pcap file. What is the number of detected packets?

✓ Correct Answer

Clear the previous alert file and comment out the old rules. Write a rule to filter packets with **Push-Ack** flags and run it against the given pcap file. What is the number of detected packets?

✓ Correct Answer

Clear the previous alert file and comment out the old rules. Create a rule to filter **UDP** packets with the same source and destination IP and run it against the given pcap file. What is the number of packets that show the same source and destination address?

7

✓ Correct Answer

Case Example - An analyst modified an existing rule successfully. Which rule option must the analyst change after the implementation?

rev

✓ Correct Answer

Task 10 Snort2 Operation Logic: Points to Remember

Points to Remember

Main Components of Snort

- Packet Decoder - Packet collector component of Snort. It collects and prepares the packets for pre-processing.
- Pre-processors - A component that arranges and modifies the packets for the detection engine.
- Detection Engine - The primary component that processes, dissects, and analyzes the packets by applying the rules.
- Logging and Alerting - Log and alert generation component.
- Outputs and Plugins - Output integration modules (i.e., alerts to syslog/mysql) and additional plugins (rule management detection plugins) support is done with this component.

Snort Rules

There are three types of rules available for Snort, which are explained below:

- Community Rules - Free ruleset under the GPLv2. Publicly accessible, no need for registration.
- Registered Rules - Free ruleset (requires registration). This ruleset contains subscriber rules with 30 30-day delay.
- Subscriber Rules (Paid) - Paid ruleset (requires subscription). This ruleset is the main ruleset and is updated twice a week (Tuesdays and Thursdays).

You can download and read more on the rules [here](#).

Note: After installing Snort2, it automatically creates the required directories and files. However, if you want to use the community or paid rules, you need to indicate each rule in Snort.conf file.

Since it is a long, all-in-one configuration file, editing it without causing misconfiguration is troublesome for some users. That is why Snort has several rule updating modules and integration tools. To sum up, never replace your configured Snort configuration files; instead, edit your configuration files manually or update your rules with additional tools and modules to prevent failures, crashes, or missing features.

- snort.conf: Main configuration file.
- local.rules: User-generated rules file.

Let's start with an overview of the main configuration file (snort.conf). `sudo gedit /etc/snort/snort.conf`.

Navigate to the "Step #1: Set the network variables" section.

This section manages the scope of the detection and rule paths.

Answer the questions below

Read the task above.

No answer needed

✓ Correct Answer

Task 11 Conclusion

In this room, we covered Snort, its definition, operation, and how to create and use rules to investigate threats.

[Download Task Files](#)

- Understanding and practising the fundamentals is crucial before creating advanced rules and using additional options.
- Do not create complex rules at once; try adding options step by step to easily notice possible syntax errors or other problems.
- Do not reinvent the wheel; use or modify/enhance it if there is a smooth rule.
- Take a backup of the configuration files before making any changes.
- Never delete a rule that works properly. Comment on it if you don't need it.
- Test newly created rules before migrating them to production.

Now, we invite you to complete a series of Snort challenges, starting from [Snort Challenge - The Basics](#).

A great way to quickly recall snort rules and commands is to download and refer to the [TryHackMe snort cheatsheet](#).

Answer the questions below

Read the task above.

No answer needed

✓ Correct Answer

Result:

Thus, Snort IDS/IPS was successfully used to detect and prevent real-time network threats using the TryHackMe platform.